HFDE-4201 EC

Crosslinkable Power Cable Insulation Compound

Overview

HFDE-4201 EC is a long-life, unfilled, low density polyethylene compound designed for medium and high voltage (5 kV and higher) power cable insulation applications. It has a very low level of contamination and bears the designation Extra Clean (EC).

HFDE-4201 EC is recommended for the insulation of power distribution cables rated up to 46 kV and can be used for transmission cables rated up to 69 kV.

Specifications

HFDE-4201 EC is designed for use in power distribution and sub-transmission cables. Cables insulated with HFDE-4201 EC, using sound commercial manufacturing practice, would be expected to meet the latest editions of the following specifications and regulations:

- ANSI/ICEA: S-94-649, S-97-682, S-93-639/NEMA WC74
- AEIC CS8
- RUS: 50-70 (U-I)
- CEA: WCWG-01, WCWG-02
- UL 1072
- IEC 60502, 60840
- CENELEC HD 620 S1/A3, HD 632 S2, HD 629 S1
- DIN VDE: 0207, DIN VDE 0276-620
- BSI BS 6622
- GB/T 12706

Physical	**Nominal Value (English)**	**Nominal Value (SI)**	**Test Method**
Density, 23°C	0.92 g/cm^3	0.92 g/cm^3	ASTM D 792
Degree of Crosslinking – Extractables	< 20 %	< 20%	ASTM D2765A
Mechanical	**Nominal Value (English)**	**Nominal Value (SI)**	**Test Method**
Tensile Strength	3000 psi	20.7 MPa	ASTM D 638
Tensile Elongation (Break)	530%	530%	ASTM D 638
Thermal	**Nominal Value (English)**	**Nominal Value (SI)**	**Test Method**
Hot Creep (392 °F/200°C)	< 100%	<100%	ICEA T-28-562
Hot Set (392 °F/200°C)	< 10%	<10%	ICEA T-28-562
Electrical	**Nominal Value (English)**	**Nominal Value (SI)**	**Test Method**
Volume Resistivity, 73°F (23°C)	>1x10^{15} ohm-cm	>1x10^{15} ohm-cm	ASTM D 257
Dielectric Strength			ASTM D 149
0.125 in (3.18 mm) Method A (Short-Time)	660 V/mil	26 kV/mm	
0.125 in (3.18 mm) Method B (Step-by-Step)	590 V/mil	23 kV/mm	
Dielectric Constant (60 Hz)	2.30	2.30	ASTM D 150
Dissipation Factor (60 Hz)	3.0E-4	3.0E-4	ASTM D 150

Additional Information

Nominal property values representing test on molded, stress-relieved slabs. Cure times were 15 minutes at 175 °C.

Extra-Clean Requirements

- HFDE-4201 EC meets the strictest standards for cleanliness established for an unfilled, crosslinkable cable insulation compound. Throughout the production process, the products is tested to ensure a high level of cleanliness. Extruded tapes are scanned by an automatic inspection system in a class 10,000 clean room. The purity data is managed using an acceptance sampling plan which ensures that the product in the shipping container meets or exceeds extra-clean standards.

- Maximum Allowable in 1.0 Kg Tape: HFDE-4201 EC
  - 125-250 μm: 2
  - 250-625 μm: 0
  - >625 μm: 0

- In addition, as HFDE-4201 EC is packaged, a continuous side stream of pellets is analyzed using an electronic pellet inspector. A review of detected contamination is incorporated into our EC quality program.

Storage

- The environment or conditions of storage greatly influences the recommended storage time. Storage under extreme conditions may affect the quality, processing, or performance of the product. Storage should be in accordance with good manufacturing practices. The recommended storage conditions, in the original unopened packages, are dry conditions with temperatures between 50ºF and 86ºF (10°C and 30°C). When stored under these conditions, the product may be used by the customer for up to one year from the date of sale or two years from the date of manufacture, whichever comes first. It is recommended that the practice of using the product on a first-in / first-out basis be established..

Extrusion	Nominal Value (English)	Nominal Value (SI)
Melt Temperature	240 to 285 °F	116 to 141 °C

Extrusion Notes

HFDE-4201 EC provides excellent surface finish and outstanding output rates over a broad range of conditions. For optimum results, melt extrusion temperatures in the range of 240 to 285 °F (116 to 140 °C) are recommended, although higher melt temperatures may be possible on certain extrusion lines with due care. Generally, a 60-40-20 screenpack is recommended. However, specific recommendations for processing conditions can be determined when the application and type of processing equipment are known.

Notes

These are typical properties only and are not to be construed as specifications. Users should confirm results by their own tests.

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